

# Which Way Does the Condition Point?

AP Statistics · Unit 2 · Topic 2.6 · B: 2026-10-26 · E: 2026-11-13 · TEACHER KEY

## CED TETHER

- **Skill 3.A** — Determine relative frequencies, proportions, or probabilities using simulation or calculations.
- **Enduring Understanding VAR-4** — The likelihood of a random event can be quantified.
- **LO VAR-4.D** — Calculate conditional probabilities. [Skill 3.A]
- **EK VAR-4.D.1** — The probability that event A will occur given that event B has occurred is called a conditional probability and denoted  $P(A | B) = P(A \cap B) / P(B)$ .
- **EK VAR-4.D.2** — The multiplication rule states that the probability that events A and B both will occur is equal to the probability that event A will occur multiplied by the probability that event B will occur, given that A has occurred. This is denoted  $P(A \cap B) = P(A) \cdot P(B | A)$ .

**Essential question:** Why can reversing the condition produce a very different probability, even when both probabilities use the same table?

**DOK-3 skill:** 4.B · **FRQ pattern:** conditional-direction-confusion-p-a-given-b-vs-b-given-a

**DOK rationale:** Part (c) weighs competing claims using the day's numerical or graphical evidence and explains a limitation or consequence; the judgment requires linked reasoning beyond a calculation.

**Do Now (first take) → Explore (video + follow-along) → Exit (finish + turn in) DOK: 1**  
→ 3 **Minutes: 45**

### Teacher does

- Show the board at the bell and collect a one-sentence first take without confirming a claim.
- Run the named video follow-along, then allow about 10 minutes for parts (a)–(c).
- Ask for evidence linked to a claim. Collect the sheet and hand-score part (c) with E/P/I.

### Students do

- Choose a claim and cite one cell or margin from the screening table.
- Complete the follow-along about conditional probability and the general multiplication rule.
- Finish (a)–(c), revise the first take in the exit line, and turn in the sheet.

### Questions to ask

- Which number or visible feature supports your claim?
- Why does the other claim go beyond that evidence?
- What would you tell someone who chose the other claim?

### Adult role

Circulate and ask students to connect their choice to evidence. Score part (c) using the three required reasoning elements; do not require dropped extension work.

## [WARN] WATCH FOR

- A choice with copied numbers but no explanation of how they support it.
- Treating a model or average as a guaranteed individual outcome.

## SCORING PART (c) — E / P / I

### Expected elements

- **reasoning-1** (required) — Rejects Mina using 45/140 or about 32.1 percent
- **reasoning-2** (required) — Uses 45 with the condition plus 95 without to form the positive group
- **reasoning-3** (required) — Explains the change in conditioning direction and denominator

|          |                                                                                                                     |
|----------|---------------------------------------------------------------------------------------------------------------------|
| <b>E</b> | Includes all three required reasoning elements above, with correct contextual evidence.                             |
| <b>P</b> | Includes at least one substantive correct reasoning link above, but omits or misstates another required element.    |
| <b>I</b> | Includes no substantive correct reasoning link above; an unsupported choice or copied number alone is insufficient. |

### Common mistakes

- Using the sensitivity 45/50 as the chance of condition after a positive result
- Dividing 45 by all 1,000 screened students instead of by the 140 positive tests
- Calling the 95 positives among students without the condition true positives
- Ignoring the 5 percent prevalence when explaining why the two conditional probabilities differ

## [STAR] TODAY'S PROBLEM — annotated

Suppose a school screens 1,000 students for a hypothetical condition. Exactly 5% have the condition. The test has 90% sensitivity: among students with the condition, 90% test positive. Its false-positive rate is 10%: among students without the condition, 10% test positive. The implied counts are shown.

**Mina:** “The test catches 90% of students who have the condition, so a positive result means a student has a 90% chance of having it.”

**Luis:** “We need to count everyone who tested positive before finding the chance of having the condition.”

Screening results (counts)

| Status       | Pos. | Neg. | Total |
|--------------|------|------|-------|
| Condition    | 45   | 5    | 50    |
| No condition | 95   | 855  | 950   |
| Total        | 140  | 860  | 1,000 |

### FIRST TAKE (5 min)

Does a positive result mean a 90% chance of having the condition? Choose a claim and cite one table count.

**Key:** Not graded. The familiar 90 percent number is intentionally tempting; the revision should use the full positive-test group.

*Rules callout “THE GIVEN EVENT SETS THE DENOMINATOR (keep on the page)” is printed on the student sheet.*

**DOK 1** (a) For  $P(\text{condition} \mid \text{positive})$ , identify the numerator and denominator counts.

**Key:** The numerator is 45 students who both have the condition and tested positive. The denominator is all  $45 + 95 = 140$  students who tested positive.

**DOK 2** (b) Find  $P(\text{positive} \mid \text{condition})$  and  $P(\text{condition} \mid \text{positive})$ . Show each count fraction and percent.

**Key:**  $P(\text{positive} \mid \text{condition}) = 45/50 = 0.90 = 90\%$ . In the other direction,  $P(\text{condition} \mid \text{positive}) = 45/140 = 9/28 \approx 0.321 = 32.1\%$ .

**DOK 3** (c) Is Mina’s claim supported? Use the positive-test counts for students with and without the condition. Explain how reversing given changes the group in the denominator. Write 3–4 sentences.

**Key:** Luis is right: Mina’s 90% uses the 50 students with the condition as the denominator. After a positive result, the relevant group is the 140 students who tested positive: 45 have the condition and 95 do not. Thus the chance of the condition given positive is  $45/140 \approx 32.1\%$ , not 90%. Reversing the given event changes the denominator and answers a different question.

### EXIT

One thing the video changed about my first take: \_\_\_\_\_